

I have a system that manages members funds. The system has 3 master accounts. The first account (Master Bank Account) maintains cash coming via imports of bank statements. The second account (Master Cash Account) maintains the cash currently held by the system. The third account (Master Fund Account) hold the total value of the fund. Each member has 2 accounts: Cash Account->cash belonging to the member, Fund Account->member's contributions to the fund. Member can fund their cash account via 2 ways: direct deposits to the master cash account which is immediately posted to the member's cash account and later cleared via matching transaction with the master bank account, or via bank import to the master bank account then reflected to the master cash account followed by posting to the member's cash account. Now, I need to develop and efficient monthly collection cycle which takes late fees application into account. Pending collection of member's contribution to the fund need to debit the member's cash account and credit the master fund account while at the same time reflecting into the member's fund account. Past the contribution cycle, late fees will apply in tiers: after 3 days x amount, after 10 days y amount, after 20 days z-amount, etc. Can you develop a best practice workflow/process flow/algorithm for the collection cycle.

A member may request a loan after x-years membership and a minimum of y-amount in his fund account. A member may borrow x-times the balance of his fund account. The funds for the loan are allocated in 2 portions: from the member's fund account and from the master fund account. A loan is considered fully repaid when the fund portion of the loan + x-% (of the requested loan amount as a settlement threshold) is repaid. A member who is delinquent cannot request a loan without admin override. The admin can also override the other loan eligibilities stated earlier. A member who is under loan repayment schedule is exempt from making monthly contributions to the fund. The member may only have x-number of active loans at any time (typically 1-2 active loans). The member can choice to exercise the option for applying or not applying a system defined grace period (in cycles, typically 1-2 cycles) at which the repayment schedule starts. The member may also choice as some point of time in the future to either fully or partially settle the loan. When partially settled, the member choice whether to rollup the schedule of skip covered repayment cycles. Repayment EMI are define by tiers: Min->Max loan amounts (example: 1K->10K EMI is 1K, 11K->30K EMI is 1.5K, etc.). The Fund also define "Fund Tiers" when it comes to allocating money associated with either loan tier. A fund tier defines the percentage of the master fund account that are available to fund loans of that loan tier. There is an additional fund tier (Emergency) that is not associated with any loan tier, used to fund emergency loan requests. Each of these fund tiers can be defined (0%->100%); in other words, fund tiers do overlap. Repayments are handled in exactly the same manner as for contribution collections, include late fees applicability. Loan requests are processed and approved/rejected by the

admin in the order they are received, exception are emergency requests are pushed to the front of the loan requests queue. Loan request are approved based on availability of funds in the associated fund tier of the corresponding loan tier. Admin may partially or fully disburse the loan amount depending on available funds for that tier. The loan is not considered active until it is fully disbursed at which point the repayment schedule is built. Disbursing a loan amount (partially or fully, will do the following: debit the master fund for the disbursed amount, reflecting that debit in the member's fund account, and crediting the member's cash account. The member may at any point request a cash out of these funds from his cash account (this is a separate process flow). Every loan request have a member guarantor. When the member misses x-number of EMI, the guarantor is notified. After y-number of missed EMI, the guarantor is liable for the remaining repayments of the loan (only the portion taken out of the master fund - excluding the member's portion + the required threshold). The loan is automatically transfer under the guarantor ownership and the borrowing member is suspended. Can you develop best practice all required workflows/process flows/algorithms to cover the above lifecycles.

As I am migrating from an older system, the membership date of a member may extend far in the past. Since contribution data may not be available, new system will start the membership with an opening balance for the member's cash and fund accounts, reflected accordingly into the master cash and fund accounts. Since these past cycle will be flagged as missed cycles leading to delinquency, I need a workflow to resolve these cycles.